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Research paper

Spatiotemporal dynamics and ecological determinants of cutaneous Leishmaniasis in Errachidia Province, southeastern Morocco: A 16-year municipality-level analysis with SARIMA forecasting

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ABSTRACT

Leishmaniasis is a vector-borne parasitic disease transmitted by female sandflies of the genus *Phlebotomus* and affects approximately 1.2 million people annually in more than 90 countries worldwide. Morocco remains one of the most affected countries in North Africa, where the disease continues to represent a major public health concern. This study aimed to analyze the spatial and temporal trends of leishmaniasis incidence across all municipalities of Errachidia Province between 2010 and 2025 and to assess the influence of ecological, demographic, and socio-economic factors on its distribution. Also, the study aimed to predict the monthly CL cases using the Seasonal Autoregressive Integrated Moving Average (SARIMA) model. Epidemiological data on parasitologically confirmed cases obtained from the Errachidia Provincial Health Delegation were processed using geographic information system (GIS) tools and statistical methods to explore spatial patterns and correlations with environmental and socio-demographic variables. The models are trained and evaluated using monthly CL cases collected from 2010 to 2025, with the optimal model selected based on Akaike Information Criterion (AIC). During 16-years study period, 7034 cases were recorded in Errachidia Province, with the highest incidence reported in 2010 (860 cases per 100,000 inhabitants). Overall, the incidence showed marked temporal fluctuations but demonstrated a general decreasing trend over the study period. A pronounced spatial heterogeneity was observed between rural municipalities (Sid Ali, Melaab, and Ferkla) and urban municipalities (Errachidia, Arfoud, and Goulmima) ($p < 0.01$). The disease was slightly more frequent in females (54.41%) than in males (45.59%), and a significant difference was observed among age groups ($p = 0.019$), with the 0–9 and 10–19-year groups being the most affected. Seasonal analysis revealed a peak incidence during winter. In addition, higher incidence was associated with low- to medium-altitude municipalities, while no significant association was observed with poverty or vulnerability indices. The SARIMA $(0,0,1)_{12}$ model demonstrated the best predictive performance. These findings highlight the heterogeneity and ecological determinants of leishmaniasis in southeastern Morocco and may support targeted surveillance and control strategies in high-risk areas.

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1. Introduction

Leishmaniasis is a vector-borne parasitic disease caused by protozoa of the genus *Leishmania* and transmitted through the bite of infected female sandflies of the genus *Phlebotomus* [1]. It affects 1.2 million people annually in more than 90 countries worldwide [2].

Leishmaniasis manifests in three main clinical forms: cutaneous leishmaniasis (CL), the most common form worldwide, visceral leishmaniasis (VL), which can be fatal if untreated, and mucocutaneous leishmaniasis (MCL) [3–5].

Morocco is among the most affected countries in North Africa, where leishmaniasis remains a major public health concern [6,7]. According to the World Health Organization, Morocco is considered one of the countries with a significant burden of cutaneous and visceral leishmaniasis, with substantial numbers of reported cases, and an estimated population at-risk of approximately 14% for CL and 10% for VL.

Despite ongoing control efforts, leishmaniasis remains one of the most frequently reported vector-borne parasitic diseases in Morocco, accounting for approximately 79.6% of cases reported nationally, which CL representing the majority of cases [8].

In Morocco, three species of Leishmania are most commonly reported, including *Leishmania major*, *L. tropica* and L. *infantum*, which cause zoonotic cutaneous leishmaniasis (ZCL, anthroponotic cutaneous leishmaniasis, and zoonotic visceral leishmaniasis, respectively [9–11]. Previous studies have shown that zoonotic cutaneous leishmaniasis caused by *L. major* is particularly widespread in the pre-Saharan regions of Morocco, with the southeastern part of the country being one of the most affected areas [12].

Several studies have highlighted the importance of environmental and socio-economic factors in shaping disease transmission. For example, zoonotic cutaneous leishmaniasis is more prevalent in pre-Saharan rural areas and villages located near palm groves, where ecological conditions favor the presence of sandfly vectors and rodent reservoirs [13,14]. In these environments, transmission intensity is influenced by environmental conditions, social and economic status, and human activities [15]. Indeed, the incidence of zoonotic cutaneous leishmaniasis is generally higher in rural areas where health services are limited and local economies rely mainly on livestock farming and agriculture [16].

The distribution of *Leishmania major* in eastern and southeastern Morocco is associated with the high density of the vector *Phlebotomus papatasi* and rodent reservoirs, which contribute to the persistence and spread of the disease in these regions [17,18].

Oasis environments in the Drâa valleys are particularly vulnerable to climatic variability and environmental change [19,20]. Climate change projections for these areas predict increasing temperatures and decreasing rainfall, which may influence the distribution of vectors and parasites and consequently affect disease transmission dynamics [21,22]. In addition, anthropogenic factors such as land-use changes and population movements may further contribute to the spread of the disease [23].

The province of Errachidia, located in the Darâa-Tafilalet region, has long been recognized as an endemic focus of zoonotic cutaneous leishmaniasis [9].

In 2010, a major outbreak was reported with more than 3400 cases, followed by a significant decline after the implementation of control measures under the national ministerial action plan [24]. However, recent data a resurgence of cases, with 105 cases reported in 2022, despite ongoing control efforts [25].

A time series refers to a sequence of observations recorded in chronological order at regular and consistent intervals, such as weeks, months, or years, with only minor deviations being acceptable. Unlike cross-sectional or general longitudinal data, time series data are characterized by their temporal dependence structure, which successive observations are often correlated, and patterns such as trends, seasonality, and cyclic variations can be identified. This distinctive structure makes time series analysis particularly suitable for studying and forecasting infectious diseases. In the context of cutaneous leishmaniasis (CL), time series modeling plays a crucial role in understanding temporal dynamics and predicting future incidence. Such predictive approaches are essential for identifying high-risk periods, anticipating potential outbreaks, and supporting timely public health interventions. Moreover, forecasting models contribute to better monitoring of disease transmission patterns, including the spread of vectors and reservoirs, as well as the influence of climatic and environmental factors. Among these approaches, Seasonal Autoregressive Integrated Moving Average (SARIMA) models have been widely used due to their ability to capture both temporal dependence and seasonal variations in epidemiological data, making them particularly appropriate for modeling monthly CL incidence in endemic regions such as Errachidia Province.

Despite the endemic nature of the disease in southeastern Morocco, comprehensive long-term analyses integrating spatial distribution and environmental determinants at the municipal level remain limited. Therefore, the present study aims to analyze the spatiotemporal distribution of leishmaniasis in Errachidia Province between 2010 and 2025, to explore the influence of environmental, demographic, and socio-economic factors on its incidence, and to develop predictive models of monthly cases using SARIMA method.

2. Materials and methods

2.1. Study area

The province of Errachidia, located in the Darâa Tafilalet region in southeastern Morocco, covers an area of 27,034 km^2 and is administratively subdivided, according to the 2015 territorial reorganization, into 29 municipalities, including 22 rural municipalities. Geographically, the province is bordered to the west by the two provinces of Tinghir and Zagora, to the south and southeast by Algeria, to the north by Midelt Province and to the northeast by Figuig Province (Fig. 1).

In terms of demographics, the province has a population of 423,055 inhabitants, with a population density of 15.64 inhabitants/km^2. Approximately 53.60% of the population lives in rural areas, reflecting the predominantly rural character of the province.

The province is characterized by a semi-arid to desert climate, with temperatures ranging from average summer temperatures reaching 31.5 °C in July and winter temperatures dropping to around 5 °C. Annual rainfall is very low (<100 mm per year) and irregularly distributed throughout the year. Additionally, the region is affected by strong seasonal winds, particularly between May and August with average speeds reaching 57.6 km/h.

2.2. Data collection

A retrospective epidemiological study was conducted using data from the national leishmaniasis surveillance and control program, specifically for Errachidia Province, the data were obtained from the Errachidia Provincial Health Delegation of the Ministry of Health and Social Protection (MSPS). The dataset covers a 16-year period (2010–2025) and initially included 7630 parasitologically confirmed cases recorded in the provincial leishmaniasis registry. It was extracted from the leishmaniasis case registry and then subjected to a process of cleaning, completion and synthesis in order to be processed in the form of descriptive and analytical statistics.

Prior to analysis, a data quality control procedure was performed. Cases recorded in municipalities located outside Errachidia Province were excluded. Records with missing information on age, sex, municipality of residence, or area of residence (urban/rural) were also removed. In addition, duplicate records were identified and excluded. After data cleaning and validation, a final dataset of 7034 confirmed

cases was retained for epidemiological, spatial, and temporal analyses.

The final database was subsequently standardized and processed for descriptive and analytical statistical analyses.

Climate data were obtained from the Ziz River Basin Agency, which manages a network of meteorological stations across the province, including Foum Zaâbel, Tadighoust, and Tazouguert. The Tadighoust station was selected as it provides the most complete and reliable historical dataset covering the entire 2010–2025 study period, with the fewest data gaps among the available stations. Climate variables included mean temperature (°C), precipitation (mm), wind speed (m/s), wind direction, and relative humidity (%); these values were applied uniformly across all municipalities included in the study. Altitude data were downloaded from the USGS Earth Explorer database, and socio-economic variables, including population density, type of settlement (urban/rural), poverty rate, and vulnerability index, were obtained from the High Commission for Planning (HCP), the official Moroccan institution responsible for national statistics (HCP, 2024).

2.3. Spatial analysis and mapping

Special analysis was performed using ArcGIS version 10.8 (ESRI, Redlands, CA, USA) to map the spatial distribution of cutaneous leishmaniasis (CL) across the study area.

Municipal administrative boundaries shapefiles were obtained from the Moroccan Geographic Information System database (www.sig-maroc.com).

In order to highlight the spatial and temporal trends in cases, the period 2010–2025 has been divided into five sub-periods, according to the following criteria: 2010 corresponds to the phase prior to the implementation of the programme to control vectors and reservoirs of *Leishmania major*; 2011–2014 represents the operational phase of the programme to control rodents and leishmaniasis vectors, accompanied by measures to improve environmental hygiene; 2015–2019 marks the decline phase, characterized by a gradual reduction in rodent and vector control measures, corresponding to the pre-pandemic period of COVID-19; 2020–2023 covers the COVID-19 pandemic phase; and 2024–2025 represents the post-pandemic phase of COVID-19.

Epidemiological and socio-environmental variables, including LC incidence, poverty, vulnerability index, sanitation access, and altitude, were integrated into the GIS database and analysed at the municipal level. The resulting maps allowed the visualization of spatial and temporal disease pattern, as well as the identification of high-risk and low-risk municipalities.

2.4. Statistical analysis

Statistical analyses were performed using Microsoft Excel 2019, Python language version 3.13 (Python Software Foundation, Wilmington, DE, USA), and IBM SPSS Statistics version 27 (IBM Corp., Armonk, NY, USA). Descriptive statistics were used to summarize the epidemiological and socio-demographic characteristics of leishmaniasis cases.

To evaluate differences in the annual number of reported cases among age groups (0–9, 10–19, 20–29, 30–39, 40–49 and ≥ 50 years), the non-parametric Kruskal–Wallis test was applied. This test was selected because the distribution of case counts did not meet the assumptions of normality required for parametric analyses.

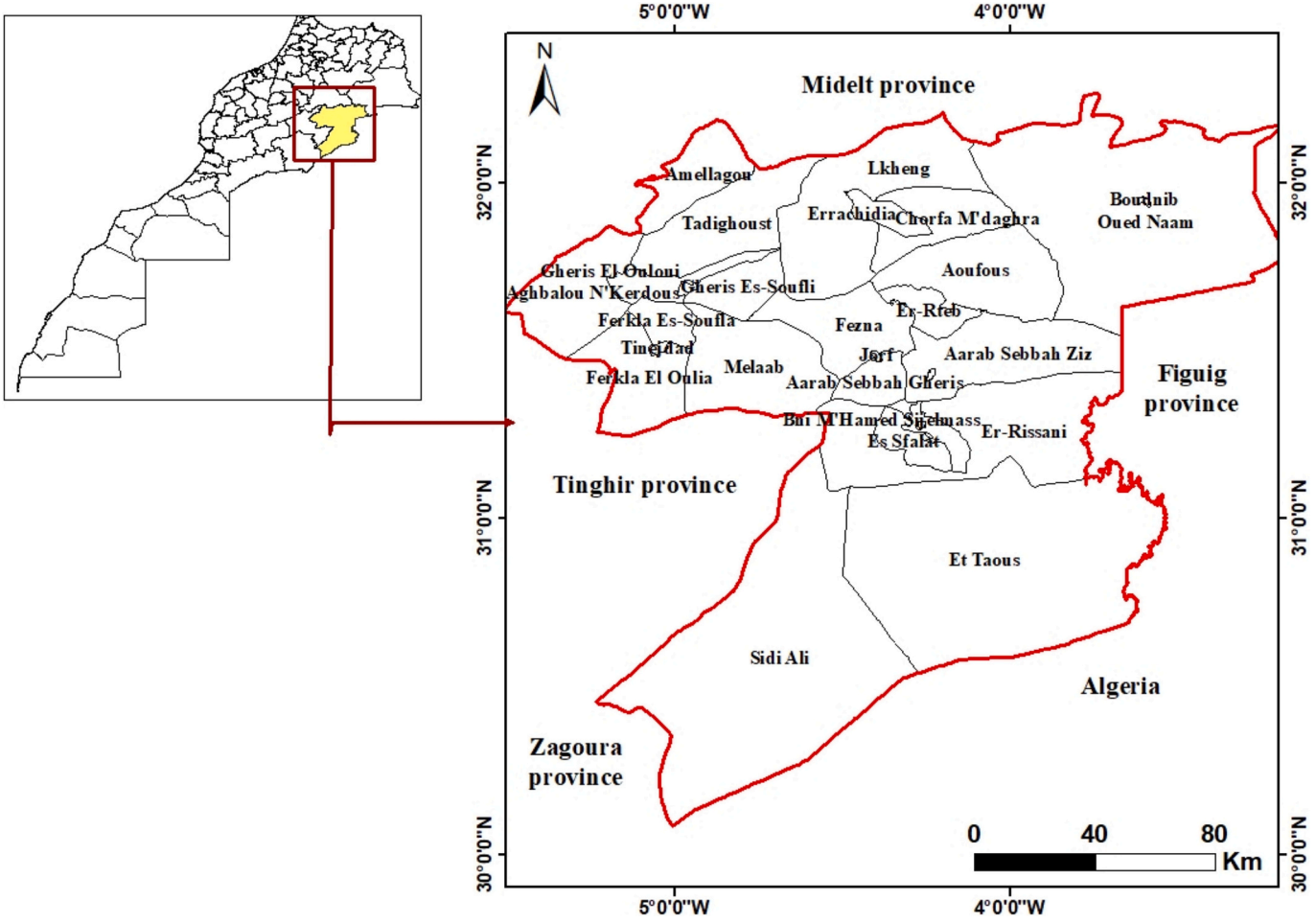


Fig. 1. Localization of Errachidia Province and the studied communes, Morocco.

The association between leishmaniasis incidence and socio-economic indicators (poverty rate, vulnerability rate and lack of access to sanitation) was assessed using Spearman's rank correlation coefficient, which is appropriate for non-normally distributed variables and monotonic relationships.

Statistical significance was set at $p \leq 0.05$.

2.5. Incidence calculation

The average cumulative incidence (IC) of leishmaniasis was calculated using the following equation:

$$IC = \frac{\text{Total number of cases recorded between 2010 and 2025}}{\text{Average population}} \times 100\,000$$

This indicator allows standardized comparison of disease occurrence across municipalities and over time. The average population was calculated separately for each municipality as the arithmetic mean of its annual population estimates over the 16-year study period (2010–2025). Annual population figures for intercensal years were estimated through linear interpolation between the known censuses (2004, 2014, and 2024) conducted by the High Commission for Planning (HCP).

2.6. Prediction methodology

Time series analysis was performed to characterize the temporal dynamics of CL incidence and to generate short-term forecasts of monthly cases. A time series is defined as a sequence of observations collected at regular time intervals, allowing the identification of trends, seasonal patterns, and temporal dependencies. In this study, monthly reported CL cases from January 2010 to December 2025 were used as a univariate time series.

The SARIMA model has been widely used to forecast future trends in the incidence of many infectious diseases [26–29]. In this study, the SARIMA model was applied to predict the monthly case number of CL, and compared with the actual cases, respectively.

The SARIMA model is generally denoted as SARIMA (p, d, q) (P, D, Q)$_s$, where p, d, and q represent the orders of the non-seasonal autoregressive, differencing, and moving average components, respectively, while P, D, and Q represent the corresponding seasonal orders. The parameters d and D represent the orders of non-seasonal and seasonal differencing required to achieve stationarity, while s represents the seasonal period, which was set to 12 months in the present study.

Accordingly, the generalized form of the *SARIM(p,d,q)* × *(P,D,Q)*s model for the monthly CL time series Y_t is mathematically expressed as follows:

$$\phi_p(\beta)\Phi_P(\beta^s)(1-\beta)^d(1-\beta^s)^D Y_t = \theta_q(\beta)\Theta_Q(\beta^s)\varepsilon_t \quad (1)$$

where Y_t denotes the monthly number of reported CL cases at time t, s is the length of the periodicity (seasonality) and ε_t is a white noise error term.

The SARIMA model was selected because of its interpretability, robustness, and suitability for relatively limited epidemiological time series. Additionally, the simple structure of the SARIMA models balances complexity and performance [30].

The Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) indicators are used to evaluate the performance of the prediction model. These are calculated According to Eqs. (2) and (3).

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \widehat{Y}_i)^2} \quad (2)$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|Y_i - \widehat{Y}_i| \quad (3)$$

In the above formulas, Y_i is the actual number of the monthly reported cases, and $\widehat{Y}_i$ is the predicted number of monthly reported cases, and N is the total number of monthly observations.

The model with the lowest RMSE and MAE values was selected as the best-performing forecasting model.

The construction of the SARIMA model followed the Box–Jenkins methodology, including model identification, parameter estimation and diagnostic checks. Firstly, an augmented Dickey-Fuller (ADF) test was performed to examine the stationary property of time series data Secondly, model parameters (p, d, q, P, D, and Q) were determined by examining the autocorrelation functions (ACF) and partial autocorrelation functions (PACF) of the transformed data. Besides, the Akaike information criterion (AIC) was used to determine the best-fi SARIMA model. Lastly, in order to test the normality of the residuals of the selected model, the Ljung-Box Q test was used to diagnose whether the residual series was white noise.

The final model was selected based on the lowest AIC value, satisfactory residual diagnostics, and predictive performance.

Residual independence was evaluated using the Ljung–Box test. A non-significant Ljung–Box test ($p > 0.05$) indicated that the residuals behaved as white noise, confirming the adequacy of the selected model.

After calibration, the selected SARIMA model was used to generate short-term forecasts of monthly CL incidence, providing estimates of future temporal trends and seasonal peaks.

3. Results

3.1. Annual incidence of leishmaniasis in Errachidia province

During the period 2010–2025, a total of 7034 cases of cutaneous leishmaniasis (CL) were reported across the 29 municipalities of Errachidia Province, representing an average cumulative incidence of 860 cases per 100,000 inhabitants at the beginning of the study period. The highest incidence was recorded in 2010, corresponding to the peak of the outbreak in the province (Fig. 2; Table 2). Following this peak, the number of reported cases decreased sharply after 2011, reaching only three cases in 2014. However, from 2015 onwards, the incidence increased again, reaching 185 cases in 2017, the number of reported cases gradually declined, reaching 11 in 2023, followed by a slight increase in 2025(Fig. 2). Overall, the temporal trend revealed marked fluctuations in CL incidence over the study period, with a general decreasing trend after the major outbreak recorded in 2010.

3.2. Temporal distribution

Regarding the temporal distribution, the monthly analysis revealed a pronounced seasonal pattern, with the majority of cases occurring during the winter months (Fig. 3). The highest numbers of cases were reported in December (1694 cases) and January (1582 cases), whereas the lowest numbers were observed during the summer months, particularly in June (50 cases) and July (64 cases). Statistical analysis confirmed a significant difference in the seasonal distribution of cases (Kruskal-Wallis test, $p < 0.01$) (Table 2).

4. Spatio-temporal distribution of leishmaniasis incidence rates in the municipalities of Errachidia

4.1. Overall cumulative incidence between 2010 and 2025

The analysis of the cumulative incidence map of cutaneous leishmaniasis (CL) in Errachidia province during the period 2010–2025 revealed that all municipalities were affected, althourgh with marked spatial heterogeneity (Fig. 4 and Table 1).

Overall, higher cumulative incidence rates were observd in rural municipalities, particularly Sidi Ali, Ferkla El Oulia, Ferkla Es-Soufla

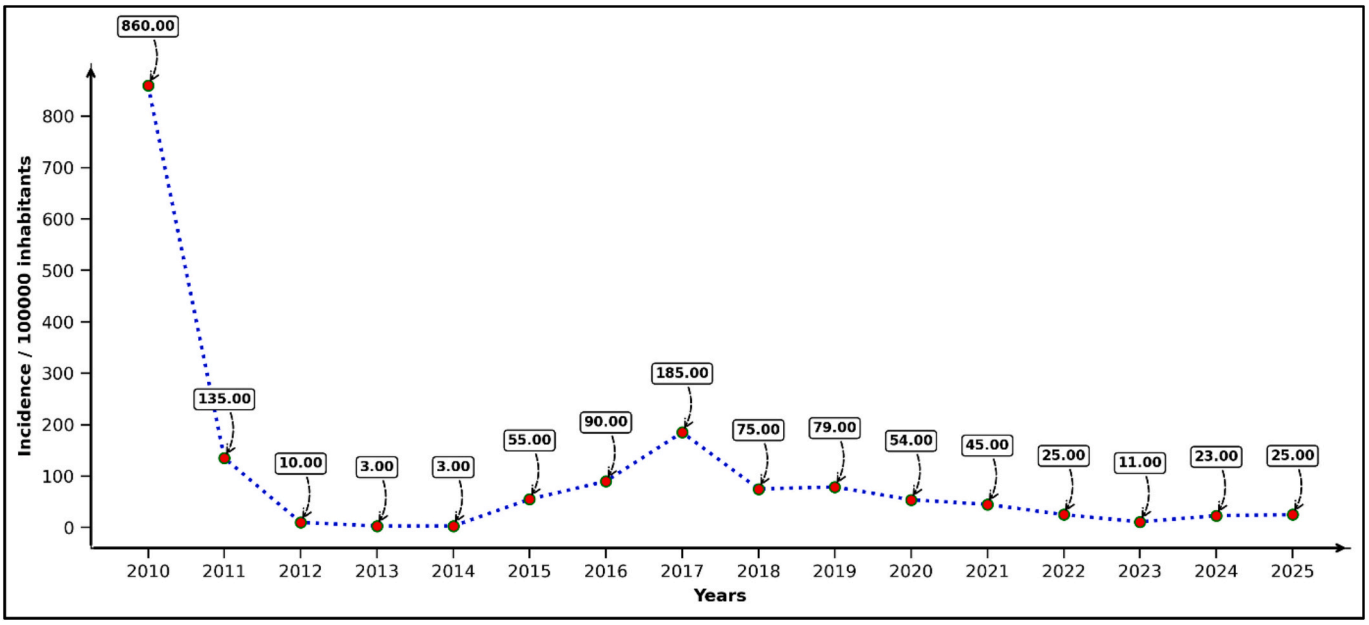


Fig. 2. Annual incidence of zoonotic cutaneous leishmaniasis in Errachidia Province, Morocco, between 2010 and 2025.

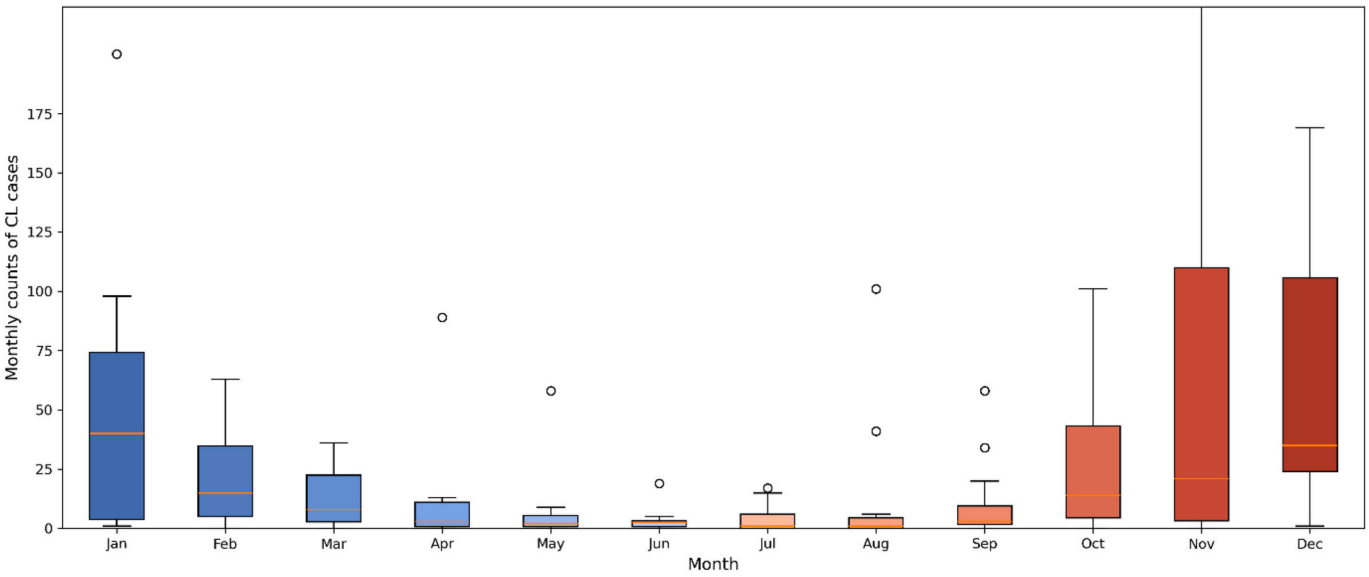


Fig. 3. Monthly distribution of cutaneous leishmaniasis cases in Errachidia Province, Morocco, during the study period (2010–2025). The boxplot represents the minimum, first quartile, median, third quartile, and maximum values.

and Melaab, compared with urban municipalities such as Errachidia, Arfoud and Er-rissani (Table 2).

The highest cumulative incidence rates were observed in municipalities located in the western and southwestern parts of the province, where the values exceeded 4000 cases per 100,000 inhabitants. In contrast, municipalities located in the northwestern, northern and northeastern areas showed considerably lower cumulative incidence rates, generally not exceeding 1000 cases per 100,000 inhabitants (Fig. 4 and Table 1).

The epidemiological evolution of CL incidence at the municipal level is illustrated in Fig. 5 and Table 1.

In 2010, all municipalities reported cases of CL, with marked geographical heterogeneity. The highest incidence rates (>1000 cases per 100,000 inhabitants) were observes in the southern and southwestern municipalities, including Sidi Ali, Es-Sfalat, Bni M'hamed Sielmass, Melaab, Ferka El Oulia and Ferkla Es-Soufla, wheras centra, northern and northeastern municipalities showed lower incidence rates.

During the period 2011–2014, the incidence declined markedly across municipalities. However, moderately high incidence rates persisted in Malaab, Ferkla and Er-rissani, where the values remained below 650 cases per 100,000 inhabitants. In contrast, Oued Naam municipality showed a slight increase, from 168 to 280 cases per 100,000 inhabitants.

Between 2015 and 2019, incidence rates increased again in several municipalities, particularly Sidi Ali, Melaab, Ghris-Es Soufli and Ferkla, where the values exceeded 1000 cases per 100,000 inhabitants. Conversely, no cases were reported in some municipalities, including Es-Sfalat, Aarab Sebbah Gheris and Amellagou, while other municipalities maintained relatively stable incidence rates below 200 cases per 100,000 inhabitants.

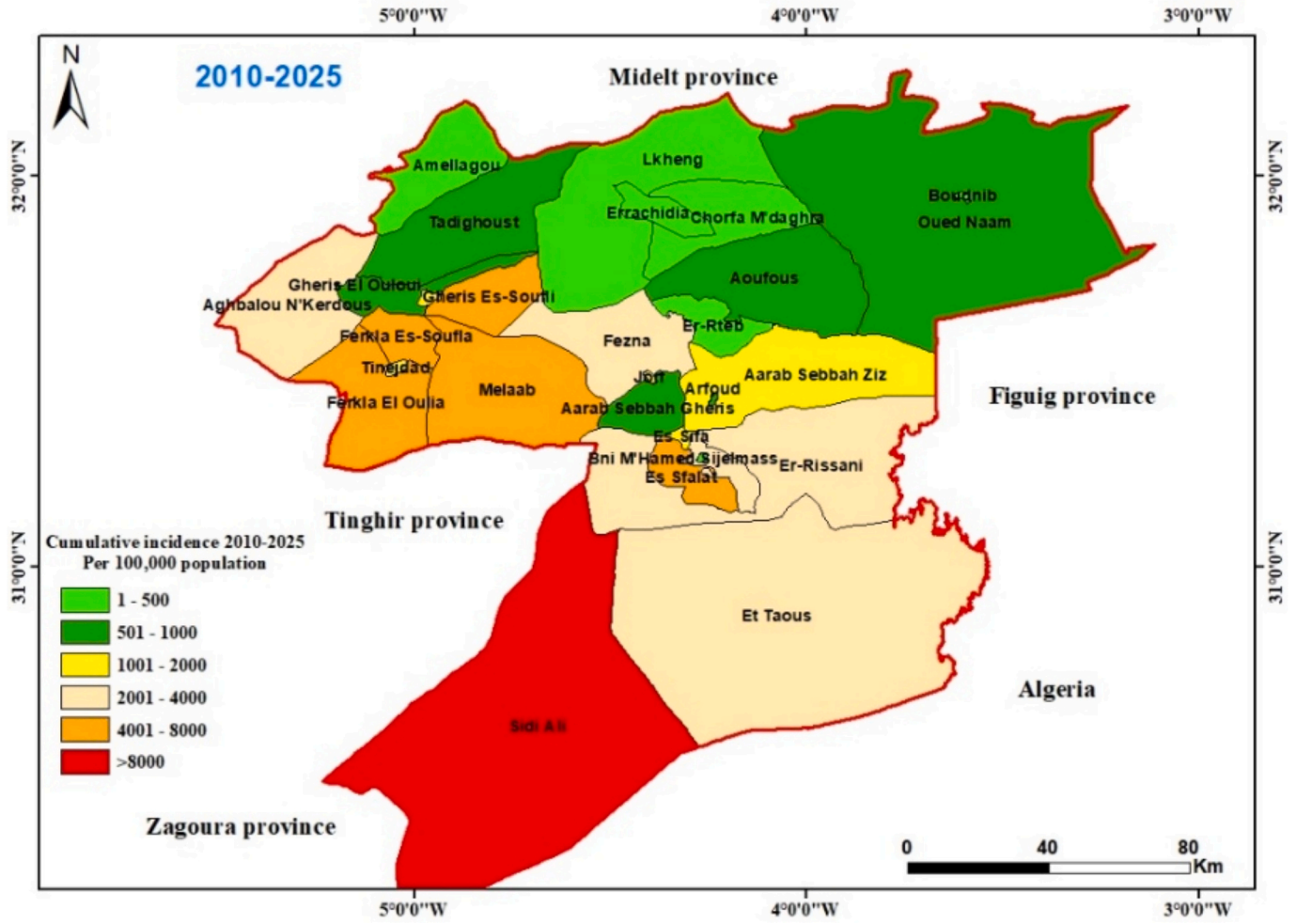


Fig. 4. Spatial distribution of cumulative incidence of cutaneous leishmaniasis in municipalities of Errachidia Province, Morocco, during the period 2010–2025.

During the period 2020–2023, the spatial distribution suggested a partial shift of the outbreak towards the southeastern part of the province. The municipality of Et-Taous, on the located near the Algerian border, recorded a sharp increase in incidence exceeding 1000 cases per 100,000 inhabitants. Similarly, Aghbalou N'Kerdous showed a notable increase, reaching 517 cases per 100,000 inhabitants. In contrast, several municipalities, including Sidi Ali, Amellagou, Tingdad, Es-Sfalat, and Aarab Sebbah Gheris, reported no cases during this period, while most other municipalities experienced a decline in incidence.

Finally, during the period 2024–2025, no municipality had an incidence exceeding 1000 cases per 100,000 inhabitants. However, Gheris Es-Soufli, Sidi Ali and Aghbalou N'Kerdous still showed relatively higher incidence rates, reaching approximately 500 cases per 100,000 inhabitants, whereas the remaining municipalities recorded very low or zero incidence rates.

4.2. Distribution by age group and gender

The distribution of CL cases according to age group and gender showed that all age groups were affected during the study period (Fig. 6; Table 2). However, the highest proportions of cases were observed among children and adolescents, particularly in the 0–9 years (31%) and 10–19 years (28.20%) age groups. The annual number of leishmaniasis cases differed significantly among age groups (Kruskal–Wallis $H = 13.54$, $df = 5$, $p = 0.019$). In contrast, no statistically significant difference was observed between genders. Overall, females represented a slightly proportion of cases, accounting for 54.62%, compared with 45.38% for males. When stratified by age groups, females predominated in most age categories, whereas males showed a slight predominance in the younger age groups (0–9 and 10–19 years).

4.3. Distribution of leishmaniasis cases according to altitude

The spatial analysis of cumulative incidence according to altitude in the municipalities of Errachidia Province revealed a clear variation in the distribution of cutaneous leishmaniasis (CL) cases (Fig. 7). Municipalities located at higher altitudes (>1200 m) generally showed very low or no recorded incidence during the study period. In contrast, municipalities located at low to medium altitudes exhibited moderate to high incidence rates, indicating a greater concentration of cases in the areas. These observations suggest a possible relationship between altitude and the geographical distribution of CL cases; however, no formal spatial statistical analysis was performed to confirm this association.

4.4. Correlation between socio-economic factors and disease incidence

The relationship between socio-economic indicators and the incidence of CL at the municipal level was assessed using Spearman's coefficient. The analysis revealed no statistically significant correlation between CL incidence and the socio-economic variables examined, including poverty rate, vulnerability index, and population density. These results suggest that, within the study area, socio-economic factors were not significantly associated with the spatial variation in CL incidence (Table 3).

Table 1
Annual distribution of cases and incidence rates of cutaneous leishmaniasis by municipality, 2010–2025.

Municipalities	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Incidence (Number of cases)															
Gheris Es-Soufli	2623 (180)	102 (07)	29 (02)	0 (00)	0 (00)	628 (43)	586 (40)	1455 (99)	900 (61)	636 (43)	0 (00)	15 (01)	0 (00)	151 (10)	455 (30)	46 (03)
Melaab	2003 (347)	565 (98)	69 (12)	0 (00)	0 (00)	865 (150)	1310 (227)	514 (89)	226 (39)	180 (31)	163 (28)	175 (30)	23 (04)	0 (00)	47 (08)	29 (05)
Tinejdad	1242 (110)	361 (32)	0 (00)	0 (00)	0 (00)	11 (01)	185 (17)	129 (12)	11 (01)	94 (09)	0 (00)	0 (00)	0 (00)	0 (00)	20 (02)	0 (00)
Arfoud	483 (140)	103 (30)	0 (00)	0 (00)	0 (00)	3 (01)	17 (05)	46 (14)	3 (01)	19 (06)	6 (02)	44 (14)	0 (00)	0 (00)	59 (19)	28 (09)
Aoufous	67 (07)	115 (12)	0 (00)	29 (03)	10 (01)	19 (02)	59 (06)	0 (00)	0 (00)	21 (02)	115 (11)	21 (02)	65 (06)	55 (05)	34 (03)	34 (03)
Goulmima	311 (51)	43 (07)	0 (00)	0 (00)	0 (00)	86 (14)	147 (24)	191 (31)	49 (08)	403 (65)	200 (32)	88 (14)	63 (10)	63 (10)	19 (03)	148 (23)
Aghbalou N'Kerdous	867 (89)	49 (05)	39 (04)	0 (00)	29 (03)	10 (01)	19 (02)	19 (02)	67 (07)	323 (34)	76 (08)	198 (21)	235 (25)	9 (01)	187 (20)	317 (34)
Lkheng	78 (11)	127 (18)	7 (01)	0 (00)	21 (03)	35 (05)	42 (06)	21 (03)	0 (00)	14 (02)	28 (04)	0 (00)	21 (03)	0 (00)	7 (01)	0 (00)
Boudnib	222 (25)	62 (07)	18 (02)	0 (00)	0 (00)	9 (01)	17 (02)	0 (00)	0 (00)	8 (01)	66 (08)	24 (03)	0 (00)	0 (00)	32 (04)	0 (00)
Jorf	366 (45)	24 (03)	0 (00)	0 (00)	0 (00)	0 (00)	24 (03)	661 (81)	139 (17)	115 (14)	98 (12)	41 (05)	0 (00)	0 (00)	8 (01)	0 (00)
Sidi Ali	3933 (87)	45 (01)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	1452 (28)	3350 (62)	900 (16)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	508 (07)
Et Taous	507 (34)	15 (01)	15 (01)	0 (00)	0 (00)	0 (00)	14 (01)	0 (00)	397 (29)	0 (00)	0 (00)	894 (69)	306 (24)	0 (00)	12 (01)	0 (00)
Es Sfalat	3323 (390)	111 (13)	34 (04)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)
Es Sifa	946 (67)	85 (06)	28 (02)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	108 (07)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)
Aarab Sebbah Gheris	587 (26)	45 (02)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)
Tadighoust	16 (01)	0 (00)	16 (01)	0 (00)	0 (00)	0 (00)	17 (01)	0 (00)	52 (03)	142 (08)	200 (11)	19 (01)	38 (02)	0 (00)	0 (00)	20 (01)
Gheris El Ouloui	474 (57)	33 (04)	25 (03)	0 (00)	0 (00)	8 (01)	8 (01)	8 (01)	8 (01)	33 (06)	136 (16)	0 (00)	0 (00)	17 (02)	0 (00)	0(00)
Ferkla Es-Soufla	4176 (516)	461 (57)	8 (01)	16 (02)	0 (00)	73 (09)	279 (34)	952 (115)	23 (28)	227 (27)	348 (41)	0 (00)	0 (00)	9 (01)	0 (00)	80 (09)
Ferkla El Oulia	2980 (637)	469 (106)	0 (00)	0 (00)	0 (00)	9 (02)	22 (05)	631 (146)	167 (39)	51 (12)	110 (26)	76 (18)	29 (07)	13 (03)	0 (00)	21 (05)
Chorfa M'daghra	14 (02)	49 (07)	14 (02)	7 (01)	21 (03)	0 (00)	7 (01)	14 (02)	0 (00)	0 (00)	99 (14)	0 (00)	0 (00)	7 (01)	7 (01)	0 (00)
Er-Rteb	30 (04)	15 (02)	7 (01)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	23 (03)	0 (00)	0 (00)	0 (00)
Oued Naam	168 (09)	131 (07)	37 (02)	111 (06)	0 (00)	0 (00)	0 (00)	0 (00)	39 (02)	20 (01)	20 (01)	0 (00)	268 (13)	0 (00)	0 (00)	0 (00)
Errachidia	30 (27)	59 (54)	2 (02)	2 (02)	2 (02)	1(01)	3 (03)	66 (64)	1 (01)	25 (25)	4 (04)	7 (07)	2 (02)	4 (04)	5 (05)	0 (00)
Fezna	696 (31)	67 (03)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	994 (45)	264 (12)	175 (08)	87 (04)	22 (01)	0 (00)	0 (00)	0 (00)	0 (00)
Er-Rissani	1038 (52)	579 (29)	20 (01)	0 (00)	0 (00)	0 (00)	0 (00)	131 (07)	37 (02)	107 (06)	35 (02)	86 (05)	101 (06)	0 (00)	16 (01)	0 (00)
Bni M'Hamed Sijelmass	2671 (389)	144 (21)	7 (01)	0 (00)	0 (00)	0 (00)	0 (00)	125 (17)	7 (01)	8 (01)	8 (01)	0 (00)	0 (00)	17 (02)	0 (00)	0 (00)
Aarab Sebbah Ziz	809 (155)	104 (20)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	73 (14)	0 (00)	31 (06)	21 (04)	0 (00)	0 (00)	32 (06)	0 (00)	32 (06)
Amellagou	40 (02)	80 (04)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)
Moulay Ali Cherif	263 (58)	41 (09)	0 (00)	0 (00)	0 (00)	0 (00)	0 (00)	35 (08)	13 (03)	22 (05)	0 (00)	0 (00)	13 (03)	0 (00)	0 (00)	0 (00)

4.5. Description of meteorological data

The monthly average distribution of meteorological variables and CL cases in Errachidia Province from January 2010 to December 2025 are shown in Fig. 8. The monthly average temperature was 20.9 ± 8.8 °C, average humidity was 39.4 ± 11.9%, total rainfall was 11.8 ± 20.1 mm, and the wind speed was 2.7 ± 0.5 m/s.

As expected for the semi-arid climate of southeastern Morocco, temperatures reached their highest values during summer and their lowest values during winter, whereas relative humidity and rainfall were generally higher during the colder seasons. However, wind speed showed relatively limited seasonal variation throughout the study period (Table 4).

4.6. Bivariate relationship between meteorological variables and cutaneous leishmaniasis

Using Pearson's correlation analysis, a significant negative correlation was found between the average monthly temperature and the number of reported CL cases ($P < 0.001$, $r = -0.491$) and the relative

Table 2
Demographic and epidemiological characteristics of patients with cutaneous leishmaniasis 2010–2025.

Variables	Total (%)	2010 n (%)	2011 n(%)	2012 n (%)	2013 n (%)	2014 n (%)	2015 n (%)	2016 n (%)	2017 n(%)	2018 n (%)	2019 n(%)	2020 n (%)	2021 n (%)	2022 n (%)	2023 n (%)	2024 n (%)	2025 n(%)	*P*-value
Annual ancidence	1653 (100)	860 (52,02)	135 (8,16)	10 (0,60)	3 (0,18)	3 (0,18)	55 (3,32)	90 (5,44)	185 (11,19)	75 (4,53)	79 (4,77)	54 (3,26)	45 (2,72)	25 (1,51)	11 (0,66)	23 (1,39)	25 (1,51)	
Gendre																		0,910
Male	3652 (45,59)	1776 (48,63)	224 (6,13)	17 (0,47)	6 (0,16)	8 (0,22)	108 (2,96)	175 (4,79)	360 (9,86)	512 (14,02)	164 (4,49)	105 (2,88)	67 (1,83)	55 (1,51)	21 (0,58)	54 (1,48)	62 (1,70)	
Female	4358 (54,41)	2344 (53,79)	316 (7,25)	26 (0,60)	12 (0,28)	5 (0,11)	118 (2,71)	209 (4,80)	404 (9,27)	440 (10,10)	186 (4,27)	111 (2,55)	71 (1,63)	50 (1,15)	24 (0,55)	42 (0,96)	46 (1,06)	
Age groupe (Years)																		0,019
0–9	2483 (31,00)	957 (38,54)	158 (6,36)	16 (0,64)	12 (0,48)	8 (0,32)	150 (6,04)	243 (9,79)	285 (11,48)	332 (13,37)	107 (4,31)	70 (2,82)	41 (1,65)	49 (1,97)	12 (0,48)	43 (1,73)	37 (1,49)	
10–19	2259 (28,20)	1317 (58,30)	165 (7,30)	12 (0,53)	3 (0,13)	3 (0,13)	34 (1,51)	65 (2,88)	169 (7,48)	273 (12,08)	77 (3,41)	53 (2,35)	36 (1,59)	18 (0,80)	11 (0,49)	23 (1,02)	25 (1,11)	
20–29	885 (11,05)	526 (59,44)	52 (5,88)	4 (0,45)	0 (0,00)	0 (0,00)	20 (2,26)	21 (2,37)	63 (7,12)	105 (11,86)	46 (5,20)	12 (1,36)	15 (1,69)	7 (0,79)	6 (0,68)	8 (0,90)	9 (1,02)	
30–39	808 (10,09)	467 (57,80)	58 (7,18)	4 (0,50)	1 (0,12)	0 (0,00)	9 (1,11)	21 (2,60)	76 (9,41)	73 (9,03)	32 (3,96)	32 (3,96)	16 (1,98)	9 (1,11)	5 (0,62)	5 (0,62)	4 (0,50)	
40–49	638 (7,97)	332 (52,04)	48 (7,52)	3 (0,47)	0 (0,00)	1 (0,16)	7 (1,10)	16 (2,51)	65 (10,19)	80 (12,54)	33 (5,17)	20 (3,13)	10 (1,57)	10 (1,57)	5 (0,78)	8 (1,25)	8 (1,25)	
>49	937 (11,70)	519 (55,39)	59 (6,30)	4 (0,43)	2 (0,21)	1 (0,11)	6 (0,64)	18 (1,92)	106 (11,31)	89 (9,50)	55 (5,87)	29 (3,09)	20 (2,13)	12 (1,28)	8 (0,85)	9 (0,96)	19 (2,03)	
Residence area																		<0,01
Urban	726 (10,46)	289 (39,81)	157 (21,63)	6 (0,83)	2 (0,28)	2 (0,28)	7 (0,96)	20 (2,75)	88 (12,12)	10 (1,38)	87 (11,98)	11 (1,52)	17 (2,34)	2 (0,28)	5 (0,69)	10 (1,38)	13 (1,79)	
Rural	6215 (89,54)	3778 (60,79)	369 (5,94)	43 (0,69)	16 (0,26)	11 (0,18)	222 (3,57)	355 (5,71)	392 (6,31)	132 (2,12)	246 (3,96)	220 (3,54)	163 (2,62)	48 (0,77)	40 (0,64)	89 (1,43)	91 (1,46)	
Season																		<0,01
Winter	4091 (51,50)	2312 (56,51)	137 (3,35)	31 (0,76)	7 (0,17)	7 (0,17)	101 (2,47)	236 (5,77)	276 (6,75)	394 (9,63)	152 (3,72)	136 (3,32)	129 (3,15)	93 (2,27)	29 (0,71)	51 (1,25)	66 (1,61)	
Spring	660 (8,31)	379 (57,42)	23 (3,48)	14 (2,12)	8 (1,31)	2 (0,30)	0 (0,00)	37 (5,61)	57 (8,64)	49 (7,42)	35 (5,30)	11 (1,67)	32 (4,85)	8 (1,21)	0 (0,00)	5 (0,76)	8 (1,21)	
Autumn	2916 (71,28)	1299 (44,55)	37 (1,27)	5 (0,17)	2 (0,07)	3 (0,10)	96 (3,29)	329 (11,28)	366 (12,55)	490 (16,80)	130 (4,46)	75 (2,67)	17 (0,58)	8 (0,27)	17 (0,58)	42 (1,44)	1 (0,03)	
Summer	277 (3,49)	137 (49,46)	17 (6,14)	2 (0,72)	2 (0,72)	2 (0,72)	3 (1,08)	4 (1,44)	59 (21,30)	17 (6,14)	16 (5,78)	10 (3,61)	5 (1,81)	1 (0,36)	0 (0,00)	2 (0,72)	30 (10,83)	
Classification																		<0,01
Autochthonous	7518 (97,97)	4107 (54,63)	536 (7,13)	41 (0,55)	10 (0,13)	8 (0,11)	227 (3,02)	580 (7,71)	429 (5,71)	554 (7,37)	457 (6,08)	73 (0,97)	168 (2,23)	82 (1,09)	41 (0,55)	95 (1,26)	104 (1,38)	
Imported	150 (1,95)	0 (0,00)	7 (4,67)	8 (5,33)	6 (4,00)	2 (1,33)	2 (1,33)	1 (0,67)	53 (35,33)	0 (0,00)	25 (16,67)	2 (1,33)	14 (9,33)	22 (14,67)	4 (2,67)	4 (2,67)	0 (0,00)	
Unknown	6 (0,08)	2 (33,33)	3 (50,00)	0 (0,00)	0 (0,00)	0 (0,00)	0 (0,00)	0 (0,00)	0 (0,00)	0 (0,00)	0 (0,00)	0 (0,00)	0 (0,00)	1 (16,67)	0 (0,00)	0 (0,00)	0 (0,00)	

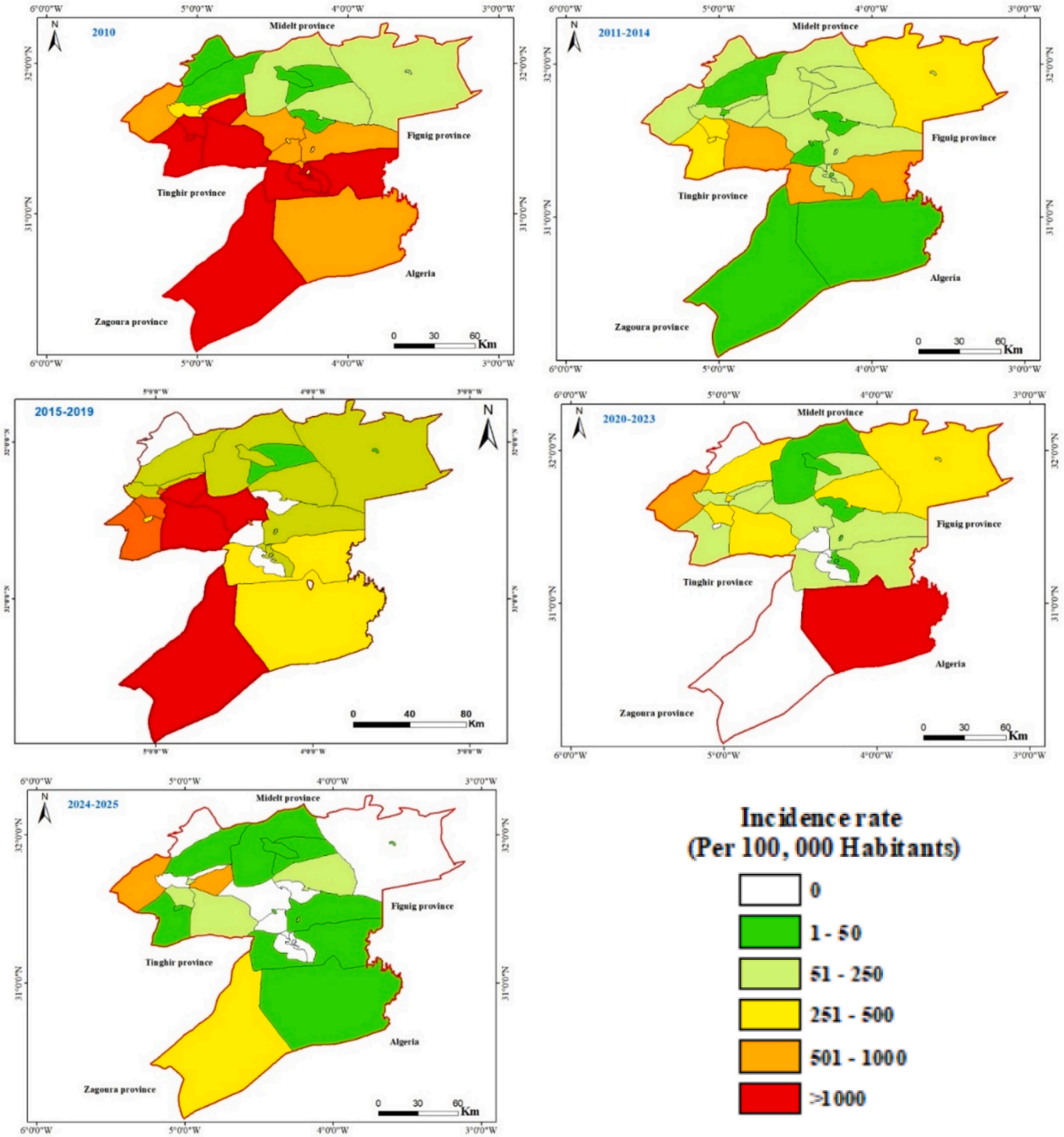


Fig. 5. Geographical distribution of incidence rate between 2010 and 2025 in communes in the province of Errachidia, Morocco.

humidity was also found to be positively associated with monthly CL cases ($P < 0.001$, $r = 0.467$). Wind speed was also negatively correlated with monthly CL cases ($P < 0.001$, $r = -0.381$). However, no significant association was observed between monthly rainfall and CL cases ($P = 0.866$, $r = 0.012$), as shown in Fig. 9.

4.7. SARIMA model

The monthly reported cases of CL from 2010 to 2025 were used as the dataset to build the prediction model. The dataset was split into training and testing sets, with 80% of the data used for training and 20% for testing.

The plot of the time series is illustrated in Fig. 10, showing the temporal evolution of monthly CL cases over the study period.

Fig. 11 represents the seasonal decomposition of the time series using the SARIMA model.

As shown in Figs. 7, 8, CL cases exhibit clear patterns of seasonality, trend, and residual components.

The autocorrelation function (ACF) and partial autocorrelation function (PACF) plots indicated the presence of temporal dependence

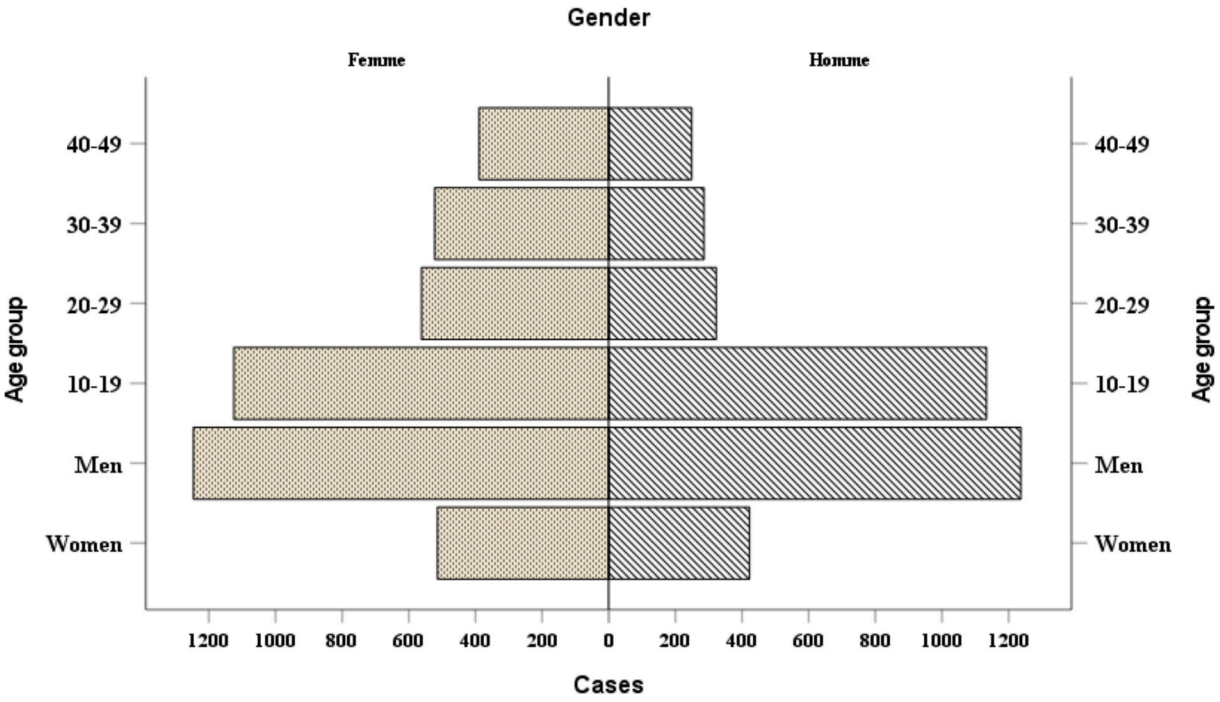


Fig. 6. Distribution of leishmaniasis cases by gender and age in Errachidia between 2010 and 2025.

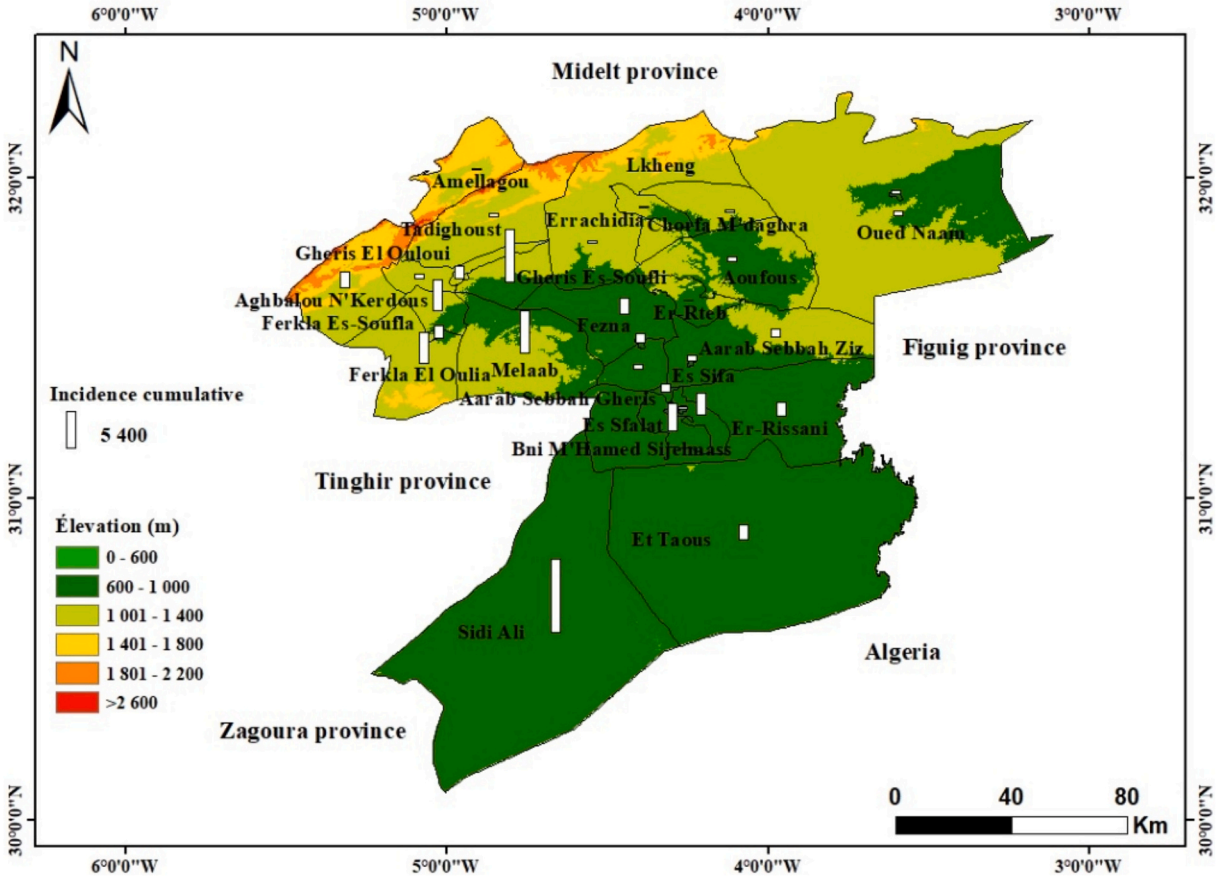


Fig. 7. Distribution of cumulative incidence according to altitude in the municipalities of Errachidia, Morocco.

and seasonal structure. As illustrated in Fig. 12, the residual ACF and PACF plots indicated that the series is not stationary. Stationarity of the series was assessed using both the Augmented Dickey–Fuller (ADF) and KPSS tests. Although, the ADF test indicated stationarity (ADF = −9.34, $p < 0.001$), the KPSS test suggested non-stationarity ($p < 0.05$), indicating the presence of residual trend components. Therefore, both first-

Table 3
Spearman's correlation between the municipal incidence of cutaneous leishmaniasis and socioeconomic factors (Poverty rate, Vulnerability rate and poor access sanitation) in Errachidia province (2010–2025).

Variable	Unit	Spearman's rho	p-value	Interpretation
Poverty rate	%	0,285	0,134	Not significant
Vulnerability rate	%	0,241	0,207	Not significant
Lack of access to sanitation	%	0,132	0,495	Not significant

$p < 0,05$ considered statistically significant

order non-seasonal differencing (d = 1) and first-order seasonal differencing (D = 1) were applied to achieve stationary data. The ACF and PACF plots of the differenced series (Fig. 13) confirmed the removal of non-stationarity. Based on these diagnostics, the SARIMA model (p, 1, q) $(P, 1, Q)_{12}$ was tentatively chosen.

An exhaustive search was conducted using a Python programming grid search method to select the optimal model based on the lowest AIC value criteria. After careful analysis, it has been determined that the SARIMA (0,1,0) $(1,1,0)_{12}$ model exhibits superior performance, as evidenced by its remarkably low AIC value of 1679.52. Moreover, the Ljung-box Q-test ($Q = 0.03$; $P = 0.86$) indicated that the model residuals were white noise, confirming the

adequacy of the selected SARIMA model.

The predictive performance of the model was evaluated using the root mean square error (RMSE) and mean absolute error (MAE), yielding values of 12.35 and 10.15, respectively. Fig. 14 illustrates the comparison between predicted and observed values, demonstrating a strong resemblance between the two. This indicates that the SARIMA (0, 1, 0) $(1, 1, 0)_{12}$ model effectively captures the temporal dynamics and seasonal variations of CL cases and provides reliable short-term forecasts.

5. Discussion

Leishmaniasis, particularly cutaneous leishmaniasis, is endemic in the southeastern region of Morocco (Drâa-Tafilalet) and more specifically in the province of Errachidia. In this study, we analyze the spatial and temporal distribution and epidemiological profile of cases reported in all municipalities of Errachidia over a 16-year period (2010–2025), with a total of 7034 cases and a peak incidence in 2010. This result is consistent with the findings of [31].

Following the implementation of an intensive cross-sectoral campaign encompassing rodent control (*Meriones shawi*) with a view to targeting *Leishmania Major* reservoir hosts, the results showed a significant reduction in the incidence rate of CL after 2010, reaching only 3 in 2014 compared to 860 per 100,000 inhabitants in 2010. However, this decrease could also be linked to the natural cycle of the epidemic, as a decline in incidence will be observed when the number of highly susceptible individuals in the population becomes too low. In addition, the immediate management of confirmed cases, public awareness, the collection and disposal of organic waste likely to form a local reservoir for sand flies, the sealing of cracks and the maintenance of hygiene have

Table 4
Seasonal average distribution of meteorological variables and monthly reported CL cases in Errachidia Province from 2010 to 2025 (mean ± SD).

Season	Mean temperature (°C)	Relative humidity (%)	Total rainfall (mm)	Wind speed (m/s)
Spring	19.34 ± 4.45	40.67 ± 6.75	11.73 ± 14.42	3.05 ± 0.33
Summer	32.00 ± 3.22	24.34 ± 5.22	5.34 ± 11.18	3.01 ± 0.25
Autumn	22.10 ± 5.81	41.84 ± 9.18	20.35 ± 28.71	2.50 ± 0.32
Winter	10.07 ± 1.85	50.64 ± 7.30	9.75 ± 19.01	2.28 ± 0.38

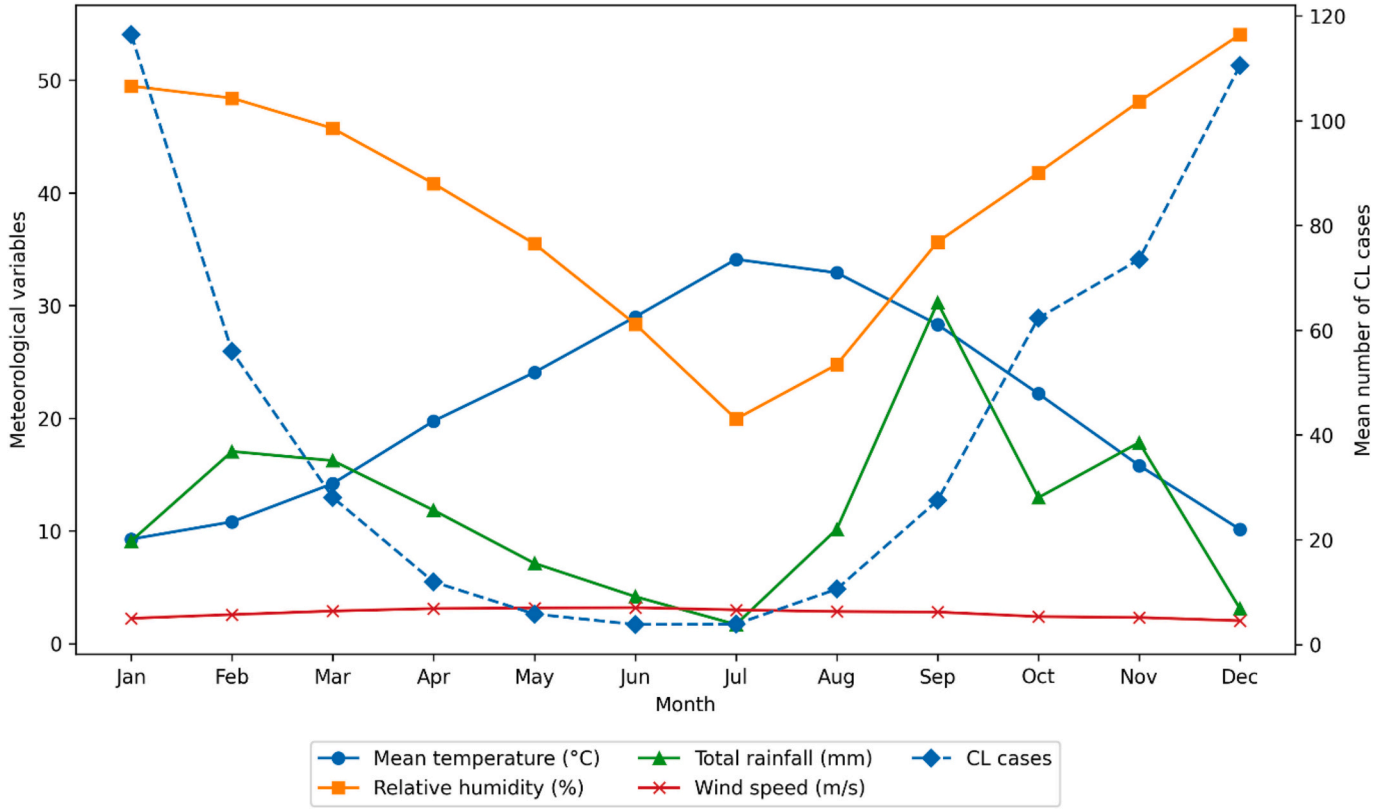


Fig. 8. Monthly distribution of meteorologica of meteorological variables and reported CL cases in Errachidia Province from 2010 to 2025.

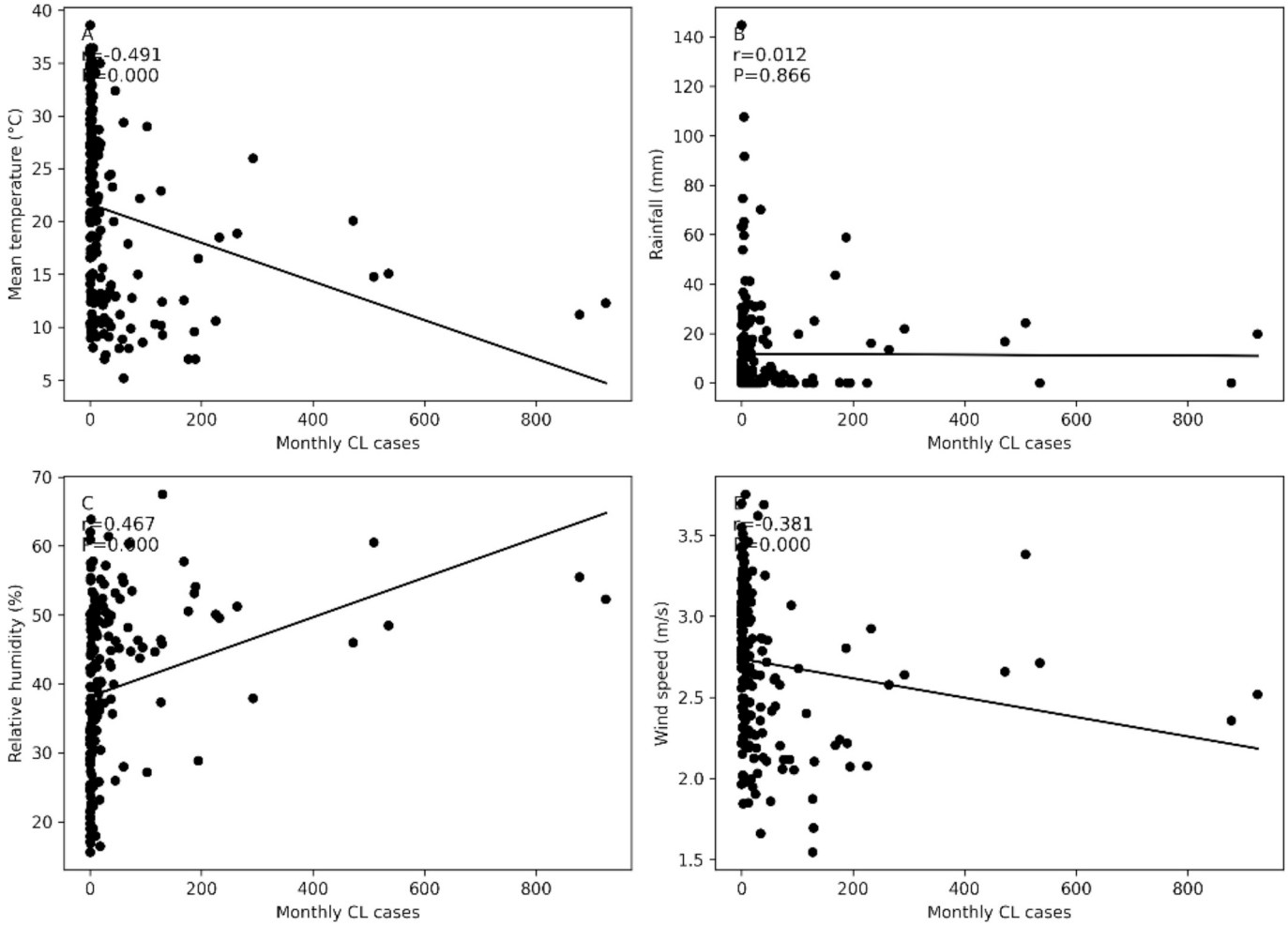


Fig. 9. Correlation between meteorological variables and monthly reported CL cases (Pearson correlation analysis)

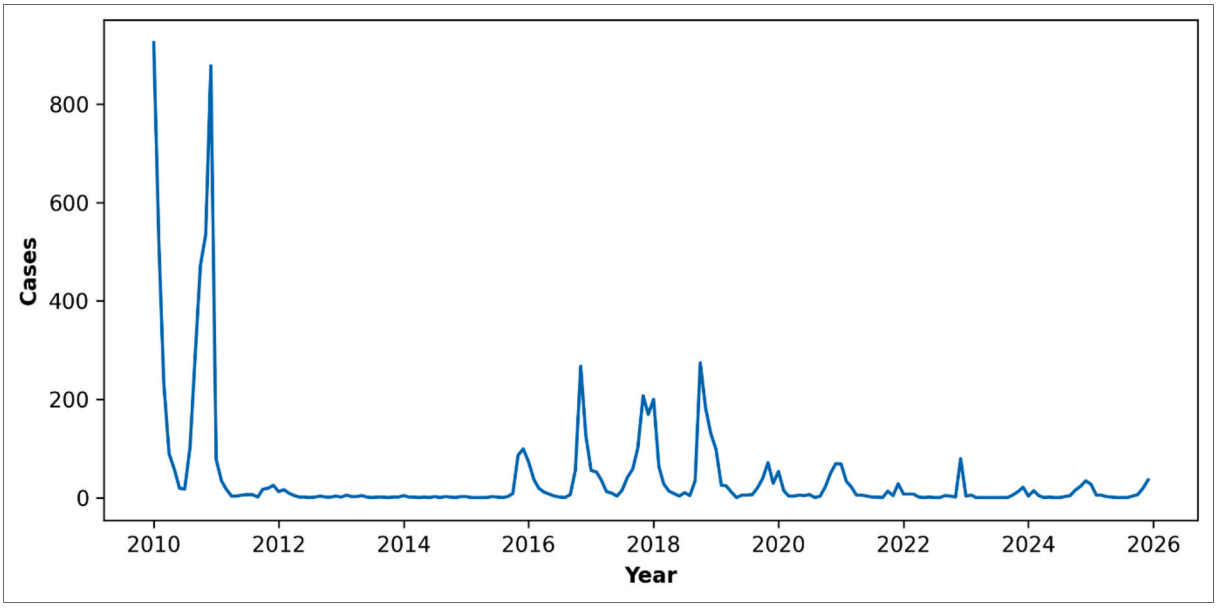


Fig. 10. Plot of the monthly cases of CL in Errachidia Province, Morocco, from 2010 to 2025.

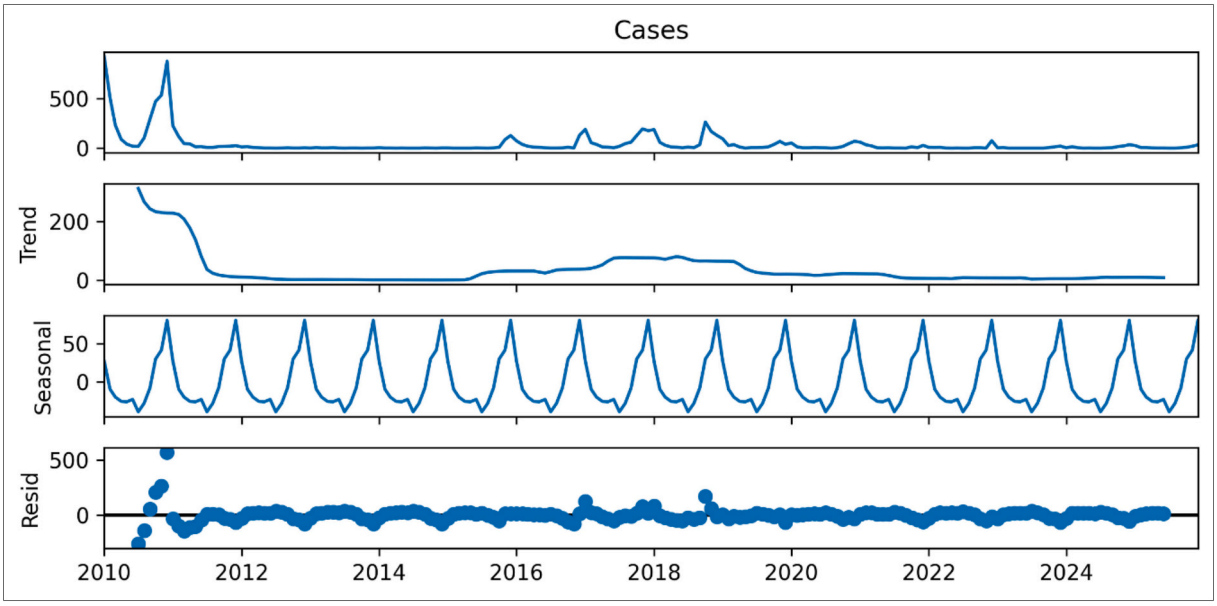


Fig. 11. Seasonal decomposition of the monthly CL cases from January 2012 to December 2025 in in Errachidia Province, Morocco.

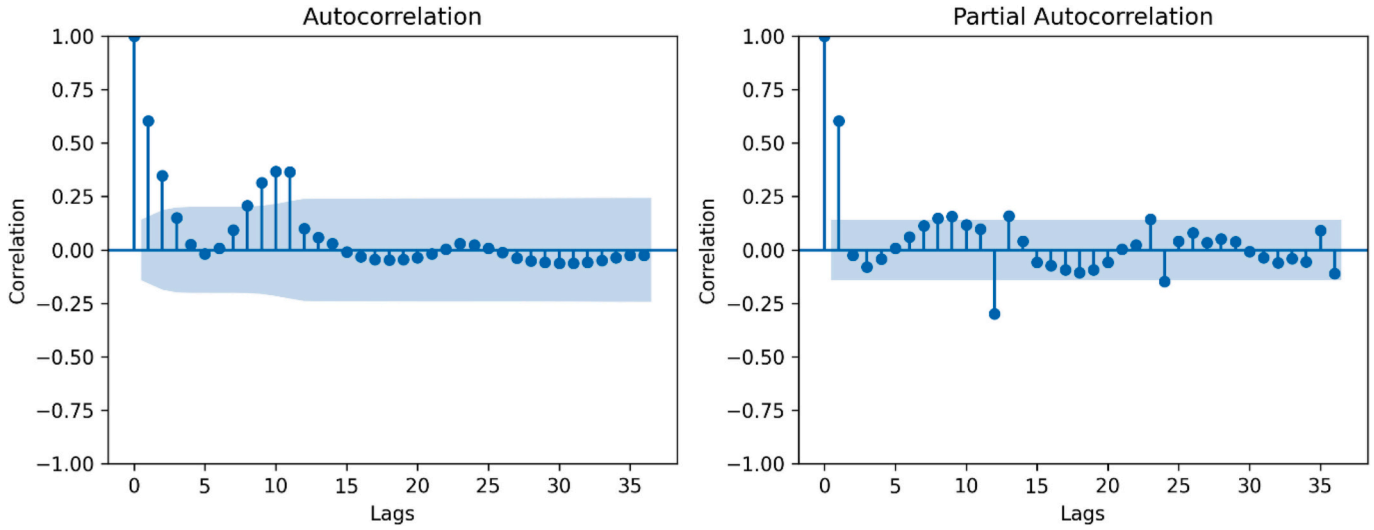


Fig. 12. Autocorrelation and partial autocorrelation plots: The original time series.

all contributed to the decline in CL cases in the province.

Between 2015 and 2019, the number of cases rose again. This increase could be explained by the relaxation of measures against rodents (*Meroine shawi*) and other actions to manage and clean up organic waste.

In the period 2020–2023, confirmed cases fell slightly again. The decline observed during this period may be hypothesised to result from a combination of factors reported in previous studies. The COVID-19 pandemic was associated with lockdown measures, intensified sanitation practices, and increased use of pesticides and disinfectants, which may have indirectly reduced human exposure to vectors and reservoir hosts [32,33]. In addition, the prolonged drought recorded during the same period may have affected the abundance of *Meriones shawi*, a known reservoir host of cutaneous leishmaniasis. Furthermore, disruptions in healthcare services and disease surveillance during the pandemic could have contributed to delays in diagnosis and underreporting of cases. However, these explanations should be considered as hypotheses based on previous literature, as the present study did not directly assess the effects of COVID-19-related interventions, drought conditions, or reservoir dynamics on disease transmission.

A slight increase in annual rainfall was recorded in Errachidia Province, from 105 mm in 2010 to 300.9 mm in 2024, coinciding with an increase in CL cases from 45 in 2023 to 105 in 2025. Based on previous studies, it may be hypothesised that higher rainfall could enhance vegetation growth and food availability, potentially favouring an increase in the population of *Meriones shawi* and thereby influencing CL transmission dynamics[26]. Nevertheless, this interpretation is based on evidence from previous studies and cannot be confirmed by the data analysed in the present work.

However, the fluctuation in confirmed cases in the province of Errachidia during this study period cannot be explained without reference to environmental factors. Studies have shown that climatic factors such as rainfall, temperature and humidity, and indeed lack of humidity and sand encroachment, could contribute to the migration of rodents to wet areas, causing a decrease in incidence[13]. However, higher temperatures could have a favourable influence on the abundance of sand flies, the main vectors of *Leishmania* spp. [34,35]. Thus, humid and warm conditions favor the multiplication of vectors, which can lead to an increase in cases after the rainy seasons [36,37].

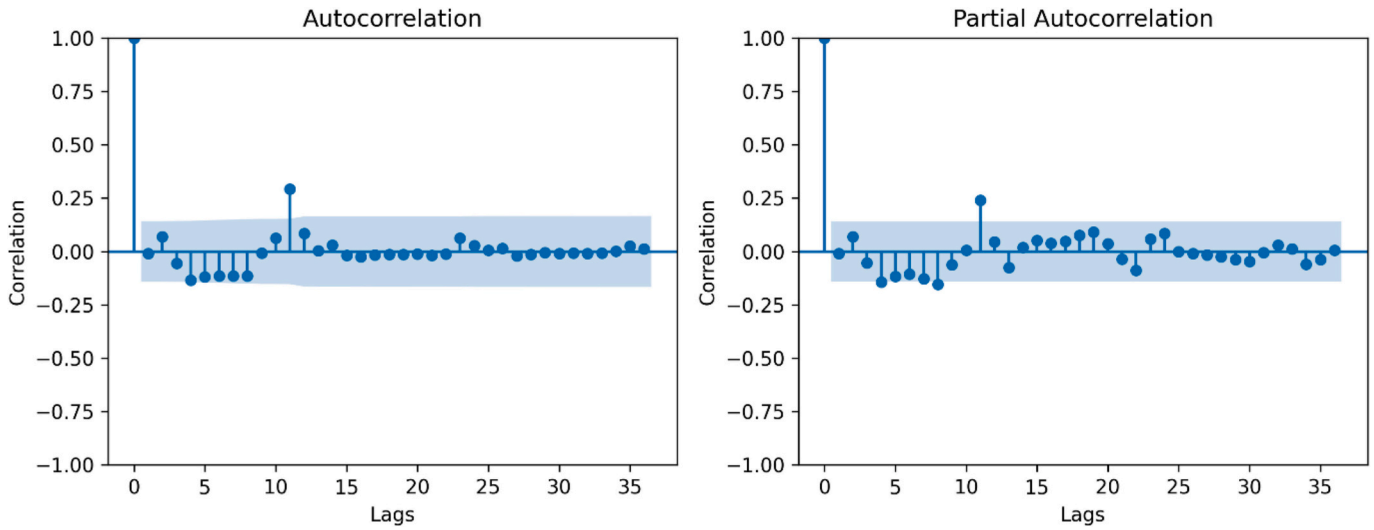


Fig. 13. Autocorrelation and partial autocorrelation plots: A first order difference and a first seasonal difference time series.

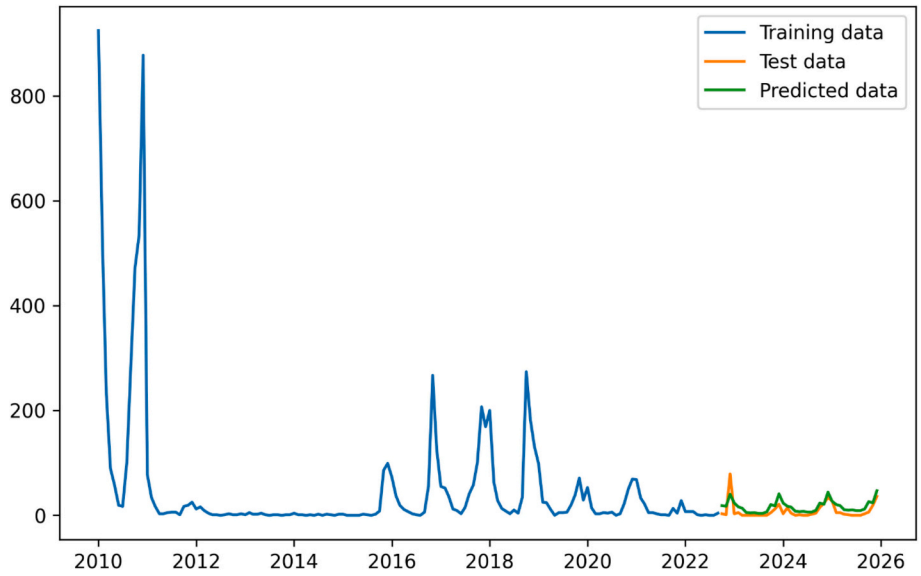


Fig. 14. Comparing predicted SARIMA and test data.

Our results show that the distribution of leishmaniasis cases appeared to vary according to geographical location. However, rural municipalities close to both palm groves and waterways (Oued Ziz, Oued Ghris and Oued Farkla) are the most affected. These results are consistent with the study conducted by [38]. Based on previous literature, it may be hypothesised that the higher soil and air humidity associated with nearby watercourses could provide favourable conditions for vector and reservoir development[39,40]. Likewise, the extensive palm groves characterising the province may offer suitable habitats and food resources for sandfly vectors, potentially favouring CL transmission [10]. Similar associations between disease risk and proximity to forests, stagnant water bodies, and rivers have been reported in other endemic settings [41]. However, these factors were not directly assessed in the present study, and their potential contribution to CL transmission in Errachidia remains hypothetical.

In south-eastern Morocco, CL is mainly caused by L. *major*, which is responsible for zoonotic cutaneous leishmaniasis [12]. In this study, *L. major* was identified as the causative agent of CL, a finding that has already been confirmed in the province of Errachidia [24].

In our study, we identified several epidemiological characteristics of the disease in the prefecture of Errachidia. Analysis of cases by age group shows that the 0–9 and 10–19 age groups are the most affected, these findings are consistent with those reported by Talbi [42]. Particularly schoolchildren and adolescents under the age of 19, which can be explained by the immaturity of the immune system and frequent exposure to sandfly bites due to their habit of playing in houses with cracked walls and close to larval breeding sites or local neighbourhood dumps, as well as travelling to school, far from their homes. [17,43].

However, the infection rate is higher among women (male-to-female ratio = 0.83:1) than among men. This is because women are more likely to consult a doctor for cosmetic reasons related to the lesions. Conversely, although men in this province are at greater risk due to their occupational activities, they tend to perceive cutaneous leishmaniasis as a neglected disease, which may lead to an underestimation of the number of cases in this population. These findings are consistent with the study by F. Talbi [42]. However, studies conducted in Iran [44] and also in the Middle East [45] show that a higher incidence is observed in men than in women.

On the other hand, the monthly distribution showed fluctuations in leishmaniasis cases, with peaks observed during the winter months (December and January). Similar seasonal patterns have been reported in previous studies [26,46–48] Based on the existing literature, it may be hypothesised that seasonal variations in case numbers could be associated with climatic factors influencing the population dynamics of sandfly vectors and reservoir hosts, as well as with the incubation period of the disease [48]. However, these factors were not directly assessed in the present study, and their contribution to the observed seasonal pattern remains hypothetical.

The distribution of incidence according to altitude appeared to vary, with low and medium altitude areas showing higher case counts. This result could be explained by ecological conditions that are more favourable to vector development, an observation that is consistent with the work carried out by Similar findings have been reported by [23], who confirmed the presence of phlebotomine biodiversity at an altitude of 800 m in the vicinity of Marrakesh. However, other studies conducted in Turkey and southern Anatolia have shown that greater phlebotomine biodiversity has been observed at altitudes of around 800 to 1000 m [49]. **It should be noted, however, that altitude was not directly analysed as an independent variable in the present study, and its potential influence on vector distribution and CL transmission in Errachidia Province remains hypothetical.**

Analysis of the influence of socio-economic factors on the incidence of cases in the municipalities of Errachidia revealed no statistically significant correlation, a finding consistent with that reported by F. Talbi in 2020 [50]. However, studies conducted in different settings have shown that the incidence of leishmaniasis is strongly associated with various climatic, demographic, and socio-economic factors [51–53]. This result confirms that leishmaniasis is multifactorial in nature, and that socio-economic factors alone are insufficient to explain disease transmission. **These explanations should therefore be considered as hypotheses drawn from previous literature, given that climatic and demographic factors were not directly examined in the present study.**

The usefulness of SARIMA models has been widely demonstrated for forecasting the incidence of several non-zoonotic infectious diseases. Previous studies have successfully applied SARIMA to predict malaria [54], cholera [55] foot and mouth disease (HFMD) [56], influenza [57], tuberculosis [58], dengue [59], and COVID-19 [60], showing that the model effectively captures temporal dependence and seasonal patterns while providing accurate short-term forecasts. These findings support the use of SARIMA as a reliable tool for epidemiological surveillance and reinforce its applicability to forecasting cutaneous leishmaniasis incidence.

Based on the comparison of AIC values, the SARIMA (0, 1, 0) (1, 1, $0)_{12}$ model was identified as the best-fit model for data. The model demonstrated good predictive performance, indicating its ability to capture both temporal dynamics and seasonal patterns in the CL time series.

The satisfactory performance of the SARIMA model highlights its suitability for forecasting monthly CL incidence in Errachidia Province. By explicitly accounting for temporal autocorrelation and annual seasonality, the model successfully reproduced the observed epidemiological dynamics while maintaining a simple and interpretable structure. These characteristics make SARIMA particularly useful for routine disease surveillance and short-term forecasting in settings where only historical surveillance data are available.

However, some limitations should be acknowledged. The SARIMA model is based solely on historical case data and assumes that the underlying temporal patterns remain relatively stable over time. Consequently, it is better suited for short-term forecasting than for long-term prospective prediction, particularly when environmental conditions, vector ecology, or public health interventions change substantially. Furthermore, the model does not explicitly incorporate exogenous variables such as climatic, environmental, or socio-demographic factors, which are known to influence CL transmission. Future studies could integrate such variables using extended models (e.g., SARIMAX) or hybrid approaches combining statistical and machine learning methods to further improve forecasting accuracy.

This spatio-temporal study of reported cases of leishmaniasis in all municipalities of Errachidia reveals observed heterogeneity in the distribution of the disease between rural and urban municipalities during this long period.

The distribution of cases by age group and gender shows high exposure among schoolchildren and adolescents, suggesting vulnerability and high mobility in high-risk environments for this group compared to other age groups.

However, analysis of the influence of socio-economic and topographical factors, mainly altitude, revealed variations in CL incidence across elevation levels, suggesting that environmental conditions may play an important role in the dynamics of leishmaniasis spread. These observations may highlight the importance of adapting control interventions to the topographical and geographical characteristics of each municipality, although this interpretation should be considered as a hypothesis based on previous literature, as these factors were not directly examined in the present study.

Finally, the study emphasises the urgent need to relaunch and strengthen the leishmaniasis control programme in the province of Errachidia, especially in the most fragile and vulnerable rural municipalities, in accordance with the recommendations of the WHO and the Minister of Health and Social Protection. Although leishmaniasis is a multifactorial vector-borne disease that is difficult to pinpoint, it is necessary to adopt an effective and rigorous strategy in order to better control and improve epidemiological surveillance and consolidate actions to combat the rodent reservoirs and vectors responsible for its spread.

6. Conclusion

This study highlights the relevance of SARIMA modeling for analyzing and forecasting the temporal dynamics of cutaneous leishmaniasis (CL) in Errachidia Province. These findings suggest that SARIMA models can serve as valuable tools for short-term forecasting and support early warning systems in endemic regions. From a public health perspective, such predictive approaches may contribute to better planning of control strategies, including timely vector control interventions and resource allocation. Future research should focus on integrating environmental and socio-demographic variables to further improve forecasting accuracy and enhance the understanding of disease dynamics.

CRediT authorship contribution statement

Omar El Alaoui: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Abdelaati El Khiat:** Writing – review & editing, Validation, Supervision, Conceptualization. **Adnane Hakem:** Visualization, Software, Methodology, Formal analysis, Data curation. **Boujemaa Ouzyou:** Resources, Investigation, Data curation. **Hajar El Omari:** Investigation, Data curation. **Benyahya Mohammed:** Resources, Data curation. **Fatima Zahrae Radi:** Data Curation, Formal Analysis, Writing – review & editing. **Omar Rhazi:** Resources. **Ilham En-niari:** Investigation. **Sanaa Mahfoud:** Investigation. **Abdelkarim Taam:** Investigation. **Fatima Zahra Talbi:** Conceptualization, Supervision, Validation, Writing – review & editing, Project Administration.

Declaration of competing interest

The authors declare no conflict of interest.

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